Informativeness vs. Complexity Conflation

Attribution reliability depends on the match between a method's assumptions and the complexity profile of the model–data system.

CROSS-CUTTING CHALLENGES: lack of ground truth · definitional ambiguity

DATA-LEVEL COMPLEXITY

exists before any model is fit · primary drivers

Collinearity

✕ suffers

KernelSHAP, marginal SHAP (Lundberg & Lee 2017)
LIME (Ribeiro et al. 2016)
PFI (Fisher et al. 2019)
Evidence: Haufe et al. (2014); Wilming et al. (2022)

✓ addresses

AEC (Salih 2024b)
Grouped / cluster-level attribution
PatternAttribution (Kindermans et al. 2018)

Feature interactions

✕ suffers

Cannot isolate the interaction term:
SHAP (per-feature), LIME
Evidence: Carmichael & Scheirer (2023)

✓ addresses

Integrated Hessians (Janizek et al. 2021)
Shapley-Taylor (Sundararajan et al. 2020)
TreeSHAP interaction values (Lundberg et al. 2020)

Suppressor effects

✕ suffers

SHAP, LIME, PFI
IG (Sundararajan et al. 2017)
LRP (Bach et al. 2015)
Evidence: Conger (1974); Wilming et al. (2022)

✓ addresses

PatternAttribution (Kindermans et al. 2018)
Haufe activation patterns (Haufe et al. 2014)

MODEL-LEVEL COMPLEXITY

introduced by the fitted model

Nonlinearity

✕ suffers

LIME, Gradient × Input
Evidence: XAI-TRIS (Clark et al. 2024)

✓ addresses

IG (Sundararajan et al. 2017)
DeepLIFT (Shrikumar et al. 2017)
NAMs (Agarwal et al. 2021)

Capacity & architecture

✕ suffers

Grad-CAM, Guided Backprop (Selvaraju et al. 2017; Springenberg et al. 2015)
fail sanity checks (Adebayo et al. 2018; Sixt et al. 2020)
Raw attention (Jain & Wallace 2019; Wiegreffe & Pinter 2019)
Additive surrogates under attention (Leemann et al. 2025)

✓ addresses

SLALOM (Leemann et al. 2025)

mismatch

Attribution failure = mismatch

a method-level assumption (below) violated by data- or model-level complexity (above)

mismatch

amplifies existing dimensions

AMPLIFIERS *(not a sixth dimension)*
Training-induced: regularization (L1/L2, dropout, early stopping); optimization & seed variability / underspecification (D'Amour et al. 2022)

MATCHING LAYER: METHOD-LEVEL ASSUMPTIONS

not part of the complexity taxonomy — what complexity is matched against

Feature independence / marginal sampling	Local linearity	Additivity (one score per feature)	Differentiability
assumed by: KernelSHAP, PFI, LIME	*assumed by:* LIME, Gradient × Input	*assumed by:* SHAP, LIME, IG	*assumed by:* IG, Grad-CAM, gradients
violated by: collinearity, suppressors	*violated by:* nonlinearity	*violated by:* interactions, attention	*violated by:* non-smooth architectures

✕ suffers = documented failures ✓ addresses = designed to be robust

Okabe–Ito colors paired with glyph + word: grayscale- and color-blind-safe